

THE ROLE OF THE CRITICAL IONIZATION VELOCITY PHENOMENA IN THE PRODUCTION OF INNER COMA COMETARY PLASMA

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Abstract—The critical velocity triggering anomalous ionization of the neutral gas by plasma flow is calculated for the model based on the lower hybride instability. It depends strongly on the plasma and gas parameters, defining the instability development of the ionized atoms beam in the counter-streaming plasma.

In particular, the possible role of the critical ionization mechanism for Halley's comet is examined. The fulfilment of both Townsend's condition for the selfsustained beam plasma discharge and Alfvén's condition for the critical velocity mechanism indicates that this mechanism may operate only within 10^4 km from the cometary nucleus and give an ion production rate close to that observed for Kohoutek's comet.

1. INTRODUCTION

Alfvén (1954, 1960) proposed that if a neutral gas and a plasma were in relative motion with a velocity exceeding a critical value V_c , then an anomalous ionization mechanism, not otherwise specified, would reduce the speed of the relative motion. This mechanism called the "critical ionization velocity phenomenon", requires an enhanced heating of electrons via some hypothetical collisionless interaction between plasma species. In the solar system this phenomenon can be of relevance in several different cases, like the solar wind interaction with a cometary atmosphere, like solar wind interaction with the Venusian atmosphere, or like corotating plasma interaction with Io torus in Jupiter's magnetosphere. In view of the visit of Halley's comet to the inner solar system, and the many fly-by missions which will explore its coma, we have developed a theory of the critical velocity phenomenon, and applied it to the cometary environment aiming to evaluate some observable quantities. Let us first discuss the value of V_c . For a lower limit Alfvén (1960) suggested

$$V_c \sqrt{\left(\frac{2e\varphi_1}{m_a}\right)} \quad (1)$$

where m_a is the neutral atom mass, φ_1 is the ionization potential and e is the electron charge. Relation (1) simply recognizes that the kinetic energy of the neutral atom in the plasma frame of reference must exceed its ionization energy $e\varphi_1$ in order to provide the energy source for the ionization. However, this energy would be available for ionization only if there is 100% efficiency in transferring energy of the neutral atoms to the electrons.

Laboratory experiments (Danielson, 1973) have shown that the enhanced gas ionization by plasma flow with the velocity above a certain critical value is accompanied by the heating of electrons and the deceleration of plasma flow.

Most theoretical models dealing with enhanced energy transfer from the flow to the electrons, invoke plasma instabilities. In principle, several types of appropriate instabilities could be excited in the situation under consideration, all of them having the same reservoir of free energy (the counter-streaming of primary plasma and neutral gas, giving rise to the secondary plasma). The access to this free energy is provided via newly ionized secondary plasma particles, and the role of the instability is to trigger the energy flow to the electrons via plasma waves. In such a case the process of enhanced ionization could be seen as high frequency discharge (with the electric fields of plasma waves playing the role of a discharge

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field). An example of a similar process is a so-called beam-plasma discharge (BPD), induced by an electron beam in a weakly ionized ionosphere, observed in active plasma experiments (Galeev *et al.*, 1976). In our case one should rather look for a BPD induced in the ion beam + neutral gas system. The comprehensive model of such BPD has to be based on appropriate instability coupled self-consistently to the condition for a self-sustaining discharge.

The previous theoretical analysis (Sherman, 1972; Raadu, 1978) revealed that the so-called "lower hybrid" instability is the most promising one for explaining the critical ionization. It is produced by counterstreaming of primary and secondary (newly ionized atoms) plasma ions in the presence of a magnetic field. At early stages of the process (prior to BPD) the "seed" secondary ions could be provided by various external sources such as photoionization, charge exchange, etc.

The character of motion of the newly ionized atoms in the crossed **E** and **B** fields associated with primary plasma flow $\mathbf{V} = cB^{-2}[\mathbf{E} \times \mathbf{B}]$ and the development of instability depend on the characteristic ionization time to the gyroperiod ratio. There are only two limiting cases:

(a) If the ionization time is shorter than the gyroperiod of ionized atoms, then we can neglect the magnetic field influence on the motion of newly ionized atoms and consider them as counterstreaming with the plasma flow on a time scale of the order of ionization time.

(b) In the opposite case the distribution of newly produced ions would be isotropic along the gyro-orbits. Their drift velocity $cB^{-2}[\mathbf{E} \times \mathbf{B}]$ would be equal to the plasma flow speed relative to the gas.

Though the plasma modes excited in both cases are the same, the character of the quasi-linear relaxation of the ionized atom velocity distribution, and thus the fraction of the ionized atom energy transferred to plasma electrons, are completely different. As a result the critical ionization velocity happens to be very different from the original estimate made by Alfven (see Equation 1).

To illustrate this we use the electron energy and number density balance in a simple form:

$$n_e \frac{dT_e}{dt} = \nu_{\text{ion}} n_e \left[\eta \frac{m_a V^2}{2} - e\varphi_1 - T_e \right] \quad (2)$$

$$\frac{dn_e}{dt} = \nu_{\text{ion}} n_e \quad (3)$$

where n_e , T_e (in energy units) are the density and the temperature of electrons, η is the fraction of the ionized atom energy $m_a V^2/2$ in a plasma frame transferred to electrons, $\nu_{\text{ion}}(T_e)$ is the ionization frequency which can be approximated by the order of magnitude as (Massey and Burhop, 1952):

$$\nu_{\text{ion}} = N \langle \sigma_{\text{ion}}(\epsilon_e) \sqrt{2\epsilon_e/m_e} \rangle \approx 10^{-2} N \sqrt{T_e} \quad (4)$$

where N is the neutral gas density, σ_{ion} is the ionization crosssection (the numerical estimate is made for hydrogen, $\sigma_{\text{ion}} = 2 \times 10^{-16} \text{ cm}^2$) and ϵ_e is the electron energy. The energy balance (2) along with the energy transferred to electrons from the ionized atoms includes also the energy spent to ionize the atom $e\varphi_1$ and to heat the new electron to the plasma electron temperature T_e . The critical ionization velocity

$$V_c = \eta^{-1/2} \sqrt{\left(\frac{2e\varphi_1}{m_a} \right)} \quad (5)$$

and the electron temperature for the plasma flow speed much above the critical velocity

$$T_e \approx \eta \frac{m_a V^2}{2} \quad (6)$$

are found from this balance equation and both of them depend strongly on the fraction of the energy transferred to electrons from ionized atoms. Thus the critical ionization velocity problem is reduced to the accurate estimate of this fraction for given plasma parameters. This will be done in the next section.

Then in the last section we will discuss the critical ionization velocity phenomena in relation to the problem of anomalous ionization of the cometary atmospheres. This problem was raised by the cometary ion observations. According to these observations the source of the cometary ions is confined to a rather small region around the nucleus (typically $\approx 10^4 \text{ km}$ for a medium comet at the distance 1 A.U. from the sun). With the typical expansion speed of the cometary gas $u \approx 1 \text{ km s}^{-1}$, this requires an ionization time scale of the order $\approx 10^4 \text{ s}$ that is much shorter than the ionization time scale for such classical processes as the photoionization by solar u.v. or the charge exchange of solar wind protons (see detailed discussion in Brandt and Mendis, 1979).

2. ION BEAM-PLASMA INSTABILITY AND ELECTRON HEATING

The linear stability analysis of an ion beam in a

plasma and the particle velocity distribution relaxation under the influence of the excited modes were extensively discussed in relation to the plasma confinement problem (Rosenbluth and Post, 1965; Galeev, 1967; Kulygin *et al.*, 1971) and solar wind-interstellar gas interaction (Wu *et al.*, 1972, 1973, 1974). General results of these studies are summarized in the monographs (Mihailosvkii, 1971; Sageev and Galeev, 1969). Here we will use the known solutions of the appropriate quasilinear equations to calculate the fraction of the ionized atoms energy spent to excite these modes. Assuming then that all these modes are absorbed finally by the tail electrons, we find the fraction of energy transferred to electrons. The rigorous proof of this assumption as well as all limitations imposed by it can be obtained by solving the non-stationary quasilinear equation for electrons. At present this was done only for the problem of plasma anomalous resistivity due to the current driven ion sound turbulence in the two limiting cases: one-dimensional relaxation in the case of the electric field along the strong magnetic field (Vekstein *et al.*, 1970) and two-dimensional relaxation in the case of the electric current across the strong magnetic field (Vekstein and Sagdeev, 1970; Vekstein *et al.*, 1971). Based on these works, we will use this assumption to find the maximum energy fraction that could be transferred to electrons after the suprathermal electron tail will be developed.

2.1 Limit of fast ionisation

Let us start first from the simplest case of the plasma flow across the magnetic field $\mathbf{B} = B\mathbf{e}_z$ with velocity $\mathbf{V} = V\mathbf{e}_x$ through such a dense gas that the following condition is satisfied:

$$\nu_{\text{ion}} \geq \Omega_+ \quad (7)$$

where $\Omega_+ = eB/M_a c$ is the cyclotron frequency of the ionized atoms. As we have mentioned above in this case we can neglect the magnetic field influence on the ionized atoms motion and consider it as one-dimensional. Thus the ionized atoms form an ion beam moving in the plasma frame along the x -axis with the velocity $\mathbf{V}_+ = -V\mathbf{e}_x$. It is known that such an ion beam excites the lower hybrid modes. The fastest growing modes propagate in the beam direction and result in the one-dimensional quasilinear relaxation of the ion beam velocity distribution. In the case of the ion production rate low with respect to the quasilinear relaxation rate we expect the form of

the newly produced ion distribution function to be stationary, i.e. it can be written in a form:

$$f_+(v_x, t) = n_+(t)F_+(v_x) \quad (8)$$

where the density $n_+(t)$ obeys the equation (3) and the function $F_+(v_x)$ has a form of a one-dimensional "plateau" (see Fig. 1a):

$$F_+(v_x) = \begin{cases} 1/V & \text{for } -V \leq v_x \leq 0 \\ 0 & \text{for } v_x > 0 \text{ and } v_x < -V \end{cases} \quad (9)$$

Here we have neglected a very small slope of the "plateau" [of the order of $\nu_{\text{ion}}/\Omega_+ \sqrt{(m_a/m_e)}$] that is needed to keep the instability development on the time scale of ionization. The amount of the ion beam energy transferred to the excited waves can be found from the energy conservation law:

$$\frac{dw}{dt} = \frac{dn_+}{dt} \left[m_a \frac{V^2}{2} - \frac{1}{2} \int_{-\infty}^{+\infty} m_a v_x^2 F_+(v_x) dv_x \right] \quad (10)$$

where W is the wave energy density. Using the expression (9) for the relaxed ion distribution function, we find the fraction of the ionized atom energy transferred to the electrons via excited waves (compare equation 10 with 2):

$$\eta = \frac{2}{3} \text{ for } \nu_{\text{ion}} \geq \Omega_+ \quad (11)$$

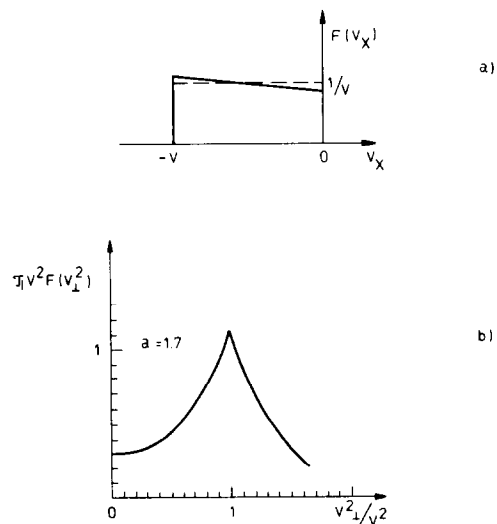


FIG. 1. MARGINALLY STABLE DISTRIBUTION FUNCTIONS FOR THE ONE-DIMENSIONAL (a) AND TWO-DIMENSIONAL (b) QUASILINEAR RELAXATION.

2.2 Limit of slow ionisation

The situation changes drastically in the opposite case when the inequality (7) is violated. The reason for that is that the fast cyclotron rotation of the ionized atoms makes their velocity distribution function isotropic in the plane perpendicular to the magnetic field lines. As a result the wave spectrum is also isotropic in the same plane. The quasilinear relaxation becomes two-dimensional, and the newly ionized atom velocity distribution spreads both to the lower and to the higher velocities perpendicular to $\vec{B}$. The velocity distribution can be represented again in the quasistationary form (compare with equation 8):

$$f_+(\mathbf{v}, t) = n_+(t) F_+(v_\perp^2) \delta(v_\parallel) \quad (12)$$

where $v_\perp$ and $v_\parallel$ are the ion velocity across and along the magnetic field. We assumed here that the neutral gas is cold and neglected its thermal velocity along the magnetic field. In principle, after the quasilinear diffusion over perpendicular velocities we should consider the energy exchange between the perpendicular and parallel motion which take place on a longer time scale of the order of cyclotron period Ω_+^{-1} .

The form of the perpendicular velocity distribution function is found in the Appendix with the help of a quasilinear theory in the limit of low ionization rate, i.e. for

$$\nu_{\text{ion}}/\Omega_+ \leq \sqrt{(m_e/m_a)} \quad (13)$$

when the quasilinear approximation is valid. The fraction of the newly ionized atoms energy transferred to electrons is numerically much smaller than in the previous case (see the Appendix):

$$\eta = 0.025. \quad (14)$$

In between the limits (7) and (13) we can expect the factor to be slowly decreasing from the value (11) to the value (14) with ionization collision frequency decreasing.

3. SELFSUSTAINED BEAM PLASMA DISCHARGE IN THE COMETARY ATMOSPHERES

The results found above will be used to estimate the time of the anomalous ionization of the cometary gas by solar wind. In view of the planned international projects to study Halley's comet in 1986 it is interesting to do the estimates for this comet. At the moment of encounter with Halley's comet the spacecraft will be at a distance of the

order of 1 a.u. from the sun. The solar wind velocity, density and temperature could be estimated then as $V_{\text{sw}} = 400 \text{ km s}^{-1}$, $n_{\text{sw}} = 5 \text{ cm}^{-3}$, $T_{e,\text{sw}} = 13 \text{ eV}$ and the interplanetary magnetic field $B_{\text{sw}} = 6 \gamma$.

We expect that the selfsustained beam-plasma discharge will take place inside the cometary atmosphere. The size "r" of the discharge region is defined by the Townsend condition for the self-sustained beam-plasma discharge (Galeev *et al.*, 1976):

$$\tau_{\text{source}} \approx \frac{1}{\nu_{\text{ion}}} + \tau_{\text{H}} < \tau_{\text{L}} \quad (15)$$

where τ_{L} is the electron life time and τ_{H} is the electron heating time. This condition ensures that each new electron has enough time to be heated and to ionize a new neutral atom. It is always satisfied in the laboratory experiments where the neutral gas density is high enough. In space and particularly in a cometary atmosphere the neutral gas density is very low at large distances from the nucleus and increases towards the latter. Therefore this condition could not be satisfied everywhere and defines, in fact, the size of the anomalous ionization region around the comet nuclei. Neglecting the heating time in comparison with the ionization time we rewrite it in terms of the cometary and solar wind parameters [using e.g. (4)]:

$$\frac{4\pi ur^2}{Q\sigma_{\text{ion}}v_e} < \frac{2r(n_e v_e)_i}{v_e(n_e v_e)_e} \quad (16)$$

where Q is the neutral atom production rate, which can be estimated to be of the order $2 \times 10^{29} \text{ atoms s}^{-1}$ for water molecules emitted by the nucleus of Halley's comet at a solar distance of 1 a.u.; u is the gas outflow speed of the order of one km s^{-1} , $\sigma_{\text{ion}} \approx 2 \times 10^{-16}$ for $T_e \leq 100 \text{ eV}$ (see Ip, 1979). The electron life time was chosen to be equal to the time of the hot discharge electrons replacement by the colder surrounding electrons. That is why the ratio of the electron fluxes inside the discharge region $(n_e v_e)_i$ and outside it $-(n_e v_e)_e$ enters this condition.

The plasma density in the discharge region can be found from the critical velocity expression (5):

$$V_c = \eta^{-1/2} \sqrt{\left(\frac{2e\phi_1}{m_a}\right)} \quad (17)$$

where parameter η is smoothly changing from the value 0.025 at the distance $r \geq 3 \times 10^4 \text{ km}$ from cometary nucleus where the condition (13) still

holds, to the value 0.67 at $r \approx 1.5 \times 10^3$ km, where the approximation of a dense gas is valid.

The solar wind is loaded by the cometary ions produced due to the photoionization by solar u.v. radiation and due to the charge exchange with the solar wind ions and can be easily decelerated to the velocity below the threshold (17). But the magnetic field lines frozen into the solar wind move faster at the flanks and thus create the field line tension inside the discharge region, where they are slowed down. The plasma flow accelerated by this tension force reaches the velocity of the order of the Alfvén velocity and thus the condition (17) defines the limiting plasma density

$$n_+ \approx \frac{\eta B^2}{8\pi e\varphi_1} \approx (1.5) \times 10^2 \text{ cm}^{-3} \quad (18)$$

where we have assumed that the magnetic field strength can reach at maximum the value $B = 50$ γ for which the solar wind dynamic pressure is of the order of magnetic pressure.

The water ion production rate can be estimated by the flux of these ions from the discharge region:

$$Q_{\text{H}_2\text{O}} \approx \pi r_d^2 V_c n_+ \approx 10^{28} \text{ ions s}^{-1}. \quad (19)$$

Here the discharge region size was put to be $r_d = 10^4$ km in agreement with Townsend's criteria and assumed critical ionization velocity behavior with the distance from the comet nucleus. This agrees with the data for Kohoutek's comet (Ip, 1979; Brandt and Mendis, 1979).

Note that our treatment is based on the assumption that the tangential discontinuity between the cometary plasma outflow region and the contaminated solar wind is well within the region of an enhanced ionization so that the solar wind flow can provide the energy source for the ionization.

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APPENDIX

CALCULATION OF THE FRACTION OF THE IONIZED ATOMS ENERGY TRANSFERRED TO ELECTRONS IN THE CASE OF STRONG MAGNETIC FIELD

In the case of a strong magnetic field when the condition (7) is violated the newly ionized atoms participate

in the cyclotron rotation around the magnetic field line. Then on the time scale of the order of the ionized atoms gyroperiod its perpendicular velocity distribution function becomes isotropic [see equation (12) in Section 2]. The quasilinear relaxation of this distribution under the influence of the excited lower hybrid modes proceeds at the longer time scale and thus should be considered as two-dimensional. The corresponding set of the quasilinear equations was derived for the case of the loss-cone instability and it is valid in our case (Galeev, 1967; Sagdeev and Galeev, 1969):

$$\frac{\partial f_+}{\partial t} = \frac{\partial}{\partial v_{\perp}} \sum_{\mathbf{k}, v_{\parallel} > \omega_{\mathbf{k}}} \frac{\omega_{\mathbf{k}}^2 e^2 |\varphi_{\mathbf{k}}|^2}{m_a^2 \sqrt{(k_{\perp}^2 v_{\perp}^2 - \omega_{\mathbf{k}}^2)}} \frac{\partial f_+}{\partial v_{\perp}} + 2\pi \frac{dn_+}{dt} \delta(v_{\perp}^2 - V^2) \delta(v_{\parallel}) \quad (\text{A.1})$$

$$\partial |\varphi_{\mathbf{k}}|^2 / \partial t = 2\gamma_{\mathbf{k}} |\varphi_{\mathbf{k}}|^2 \quad (\text{A.2})$$

where

$$\gamma_{\mathbf{k}}^+ = \frac{2\pi\omega_{\mathbf{k}}^2 + \omega_{\mathbf{k}}^2}{k^3 n_+} \int_{-\infty}^{+\infty} dv_{\parallel} \int_{\omega/k_{\perp}}^{\infty} dv_{\perp} \frac{\partial f_+}{\partial v_{\perp}} \left[v_{\perp}^2 - \frac{\omega_{\mathbf{k}}^2}{k_{\perp}^2} \right]^{-1/2} / \left[1 + \frac{\omega_{pe}^2}{\Omega_e^2} + \frac{2\pi\omega_{p+}^2}{k^2 n_+} \int_{-\infty}^{+\infty} dv_{\parallel} f_+(v_{\perp}^2 = 0, v_{\parallel}) \right]$$

$$\omega_{\mathbf{k}}^2 = (\omega_{p+}^2 + \omega_{pe}^2 k_{\parallel}^2 / k^2)$$

$$/ \left[1 + \frac{\omega_{pe}^2}{\Omega_e^2} + \frac{2\pi\omega_{p+}^2}{k^2 n_+} \int_{-\infty}^{+\infty} dv_{\parallel} f_+(v_{\perp}^2 = 0, v_{\parallel}) \right].$$

Here $k^2 |\varphi_{\mathbf{k}}|^2 / 8\pi$ is the electric field energy density for the mode with the wave vector $\mathbf{k}$ and the frequency $\omega_{\mathbf{k}}$, $\omega_{pj} = \sqrt{4\pi e^2 n_j / m_j}$ is the plasma frequency and $\Omega_j = eB/m_j c$ is the cyclotron frequency of the species j . The newly ionized atom moves along the cyclotron orbit with the velocity V equal to the gas flow velocity in the plasma coordinate system. The frequency $\omega_{\mathbf{k}}$ and the growth rate $\gamma_{\mathbf{k}}$ were calculated by Rosenbluth and Post (1965) and could be found also in the Mikhailovskii's book (1971). It is known that the two-dimensional quasilinear spread of the monoenergetic distribution of ionized atoms suppress all the excited modes, except those with small phase velocities ($\omega/k \ll V$) (see Fig. 1b). This simplifies equation (19) and makes if possible to find its solution

$$f_+(v_{\parallel}, v_{\perp}^2, t) = \frac{n_+(t)}{\pi V^2} \psi \left(\frac{v_{\perp}^2}{V^2} \right) \delta(v_{\parallel}) \quad (\text{A.3})$$

where

$$\psi(w) = (5a^2/4)w^{3/4} \left\{ I_{3/5}(aw^{5/4}) K_{3/5}(a) \theta(1-w) + K_{3/5}(aw^{5/4}) + I_{3/5}(a) \theta(w-1) + 2K_{3/5}(aw^{5/4}) K_{3/5}(a) / \Gamma(2/5) \Gamma(3/5) \right\} \quad (\text{A.4})$$

†In fact as in the previous case the growth rate ($\gamma_{\mathbf{k}}^+ + \gamma_{\mathbf{k}}^{\pm}$) is positive to account for the relaxation of the slowly new coming ionized atoms.

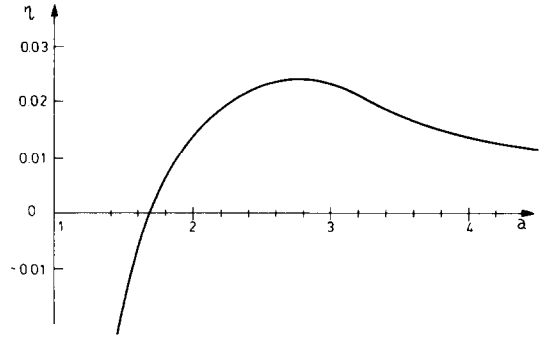


FIG. 2. FRACTION OF THE IONIZED ATOM ENERGY $m_a V^2/2$ TRANSFERRED TO THE TAIL ELECTRONS AS A FUNCTION OF THE QUASILINEAR DIFFUSION TIME/IONIZATION TIME RATIO.

Here $\theta(w)$ is the step function [$\theta(w) = 1$, for $w \geq 0$ and $\theta(w) = 0$ for $w < 0$], $I_{3/5}$ and $K_{3/5}$ are the modified Bessel functions. $\Gamma(x)$ is the gamma function, and parameter a can be expressed in term of the ratio of the quasilinear diffusion time to the ionization time:

$$\frac{25}{16} a^2 = \nu_{\text{ion}} / \sum_{\mathbf{k}} \frac{4\omega_{\mathbf{k}}^2 e^2 |\varphi_{\mathbf{k}}|^2}{kV m_a^2 V^4}. \quad (\text{A.5})$$

Multiplying the equation (A.1) by the ionized atom energy $m_a v_{\perp}^2/2$ and integrating the result over the perpendicular velocities we obtain the energy conservation law in the form

$$2 \sum_{\mathbf{k}} \gamma_{\mathbf{k}} W_{\mathbf{k}} = \nu_{\text{ion}} n_e m_a V^2 / 2 \cdot \left[1 - \int_0^{\infty} w \psi(w) dw \right] \quad (\text{A.6})$$

where $W_{\mathbf{k}}$ is the mode energy density defined as

$$W_{\mathbf{k}} = \frac{k^2 |\varphi_{\mathbf{k}}|^2}{4\pi} \left[1 + \frac{\omega_{pe}^2}{\Omega_e^2} + \frac{\omega_{p+}^2}{k^2 V^2} \Psi(0) \right]. \quad (\text{A.7})$$

Using the expression for the growth rate given above we can obtain the fraction of the ionized atoms energy spent to excite the modes:

$$\eta = \left[1 - \int_0^{\infty} w \psi(w) dw \right] - \frac{16}{25a^2} \int_0^{\infty} dw \frac{d\psi}{dw} w^{-1/2}. \quad (\text{A.8})$$

It is shown at Fig. 2 as a function of parameter “ a ”. We see that it is exactly zero for $a = 1.7$ since the growth rate is also zero at this point. In the absence of the tail electrons the marginally stable distribution function ψ should be close to that described by equation (A.4) with $a = 1.7$. However, in the presence of the tail electrons (as we have assumed here) the rate of plasma mode excitations can be compensated by their absorption by electrons and plasma will still be marginally stable.† Then the fraction of the energy transferred to the electrons can

reach the maximum value at $a = 2.8$:

$$\eta = 0.025. \quad (\text{A.9})$$

neglected (see details in Galeev, 1965), i.e.

$$\gamma_k^+ \gg \sum_l \frac{c^2 k^4 |\varphi_l|^2}{B^2 \omega_k}. \quad (\text{A.10})$$

Finally the range of validity of this estimate given by equation (13) can be found from the condition that nonlinear resonance three wave coupling could be

Here the level of turbulence is defined by equation (A.5) for given $a = 2.8$.